MAHIMA SAHU

AI Engineer • Agentic AI • MCP • LLM Systems • Python • Azure • Backend Engineering

mahimasahu6697@gmail.com | +91 | github.com/Boredletsgo | linkedin.com/in/mahima-sahu

PROFESSIONAL SUMMARY

AI Engineer with ~2 years of hands-on experience delivering agentic AI systems, LLM-powered automation pipelines, and enterprise test intelligence solutions inside Microsoft's engineering ecosystem — across Azure DevOps and Xbox. Promoted from Associate Software Engineer to Associate AI Engineer within the same engagement for demonstrated impact. Proven ability to take ambiguous business problems and ship production-grade AI solutions: from MCP server architecture and plugin-based LLM control planes to Playwright automation frameworks for large-scale legacy .NET applications. Skilled in Python, C#, Azure OpenAI, LangChain, RAG, and REST API-driven distributed systems.

TECHNICAL SKILLS

Programming	Python, C#, SQL, C
AI / GenAI	Azure OpenAI, LangChain, Prompt Engineering, LLM Orchestration, Agentic Systems, MCP, RAG, Embeddings, Vector Databases, NLP
Test Automation	Playwright, TDD, Test Plan Design, Coverage Matrix Design, Azure DevOps Test Plans & Suites
Backend & APIs	Flask, FastAPI, ASP.NET Core, REST APIs, SQLAlchemy, .NET Framework, Plugin Architecture
Cloud & DevOps	Azure App Service, Azure Storage, Azure SQL, Azure DevOps, Azure AI Search, CI/CD, Docker, Git
Systems	Distributed Systems, Workflow Orchestration, System Design, Performance Optimization, Security-by-construction

PROFESSIONAL EXPERIENCE

Tech Mahindra — Client: Microsoft Xbox *Aug 2026 – Present*

Associate AI Engineer | Promoted from Associate Software Engineer

Stack: Python, C#, .NET Framework, Playwright, Azure OpenAI, LLM Agents, Azure DevOps

- Leading AI-driven automation strategy for large-scale legacy Xbox .NET applications — assessing architecture, testing patterns, and dev workflows to identify and prioritise the highest-ROI automation opportunities across the Xbox engineering org.
- Designing and prototyping Playwright end-to-end test frameworks and coverage matrices for complex enterprise Xbox applications, establishing modern testing foundations on legacy codebases.
- Translating Xbox engineering requirements into LLM-backed automation proposals and proof-of-concepts — owning the full cycle from business problem to working prototype.
- Designing and validating AI-powered automation solutions that reduce manual QA effort, improve release velocity, and accelerate operational workflows at enterprise scale.

Tech Mahindra — Client: Microsoft *Nov 2024 – Aug 2026*

Associate Software Engineer

Stack: Python, C# .NET Core, MCP, LLM Agents, Azure DevOps REST APIs, Plugin Architecture

- Built a plugin-based AI control plane that converts natural-language engineering specs into federated Azure DevOps test plans — eliminating hours of manual test authoring per release cycle and measurably accelerating team productivity.
- Designed a dual-surface architecture pairing a conversational LLM agent (custom agent profile, skill manifests, MCP STDIO server) with a deterministic headless execution engine — keeping LLM output on the user side of the trust boundary.

- Authored a test-engine-agnostic plugin contract bounding onboarding of any future execution tool to a single new plugin, plus an org-wide agent-customization blueprint covering agent profiles, skill schemas, and MCP registration.
- Shipped a production GitHub CLI AI agent and C# .NET Core RESTful APIs with TDD and Git-based CI/CD pipelines; drove end-to-end traceability across 100+ test scenarios.
- Enforced security-by-construction: deny-by-default egress, policy-as-code authorization allowlists, SSO-only credentials, redacted append-only audit trails, and confirmation guards on every mutating operation.
- Reduced engineer onboarding effort by ~40% through automated test plan scaffolding and standardized documentation workflows.

PROJECTS

Story_Scout — Multi-Agent RAG Recommendation Platform 2025–Present

- Built a production-grade multi-agent book recommendation system using LangGraph, RAG, vector databases, and memory systems for personalized, context-aware reasoning.
- Designed agent orchestration and retrieval pipelines to generate recommendations grounded in user history rather than static filtering.

UpRate — AI Assessment & Mentorship Platform *2024–Present*

- Designed and developed an AI-powered assessment and mentorship platform with personalized candidate evaluation using Azure OpenAI and Python/Flask.
- Built resume intelligence workflows for automated profile generation, skill extraction, and candidate scoring using LangChain and SQLAlchemy.
- Developed configurable evaluation pipelines with automated scoring, feedback generation, and REST APIs deployed on Azure App Service.

AI Log Analyzer *2024*

- Built an AI-assisted log analysis platform using Python and LangChain to detect anomalies and accelerate root-cause investigation across test deployments.
- Implemented automated log parsing, summarization, and intelligent classification workflows reducing mean time to diagnosis.

MENTLAZA — Emotion Recognition Platform *2023*

- Developed a real-time emotion recognition system using DeepFace and OpenCV, achieving 96% classification accuracy across 7 emotion categories.

SENTI — Voice Sentiment Analysis *2023*

- Built a voice sentiment analysis system combining speech recognition, NLP, and VADER to classify emotional tone from live audio input.

RESEARCH & PUBLICATIONS

Addressing the Challenges of Handling Missing Data in Data Science Applications — IJIRIS, 2023 | DOI: 10.26562/ijiris.2023.v0903.05

Sentiment Analysis of Incoming Voice Calls — IJISRT, Jul 2024

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102)
- Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103)
- Microsoft Certified: Azure Data Scientist Associate (DP-100)
- Microsoft Certified: Security, Compliance & Identity Fundamentals (SC-900)
- GitHub Copilot Certification (GH-300)

EDUCATION

Birla Institute of Technology and Science, Pilani *2026-2028*

M.Tech Artifical Intelligence and Machine Learning

Sri Sairam College of Engineering, VTU *Graduated 2024*
B.E. Computer Science and Engineering (Honors) | CGPA: 8.43 / 10